# INTERRUPTION TIMING UNDER HIDDEN INFORMATION: SHARP BOUNDS ON CONFLICTING STATES

Abstract. A defender can spend a resource before an irreversible deadline or retain it for a later threat whose target becomes observable. We study this trade-off in a finite model motivated by a graveyard-removal decision in Yu-Gi-Oh! Each hidden state specifies urgent targets, a pool of later targets, and a bound on the number of distinct late requests. Every removal consumes one unit from a shared budget. We derive the exact minimum budget and prove that every family infeasible at budget $k$ contains an infeasible subfamily of at most $k+1$ states. If every state has late demand at least $d$ after the canonical early action, the bound improves to $1 + \max\{k - d, 0\}$. Both bounds are sharp. A three-state example is pairwise feasible but jointly infeasible and shows that the budget function can violate submodularity and LP-type locality. When one action can remove several targets, minimal obstructions can be arbitrarily large even at budget one. These results show how the granularity of a resource controls the order of conflicts under hidden information.

## 1. Introduction

An interruption has value both for the threat it can stop immediately and for the threat it may stop after an opponent reveals a continuation. Under hidden information, the same public position can support continuations that favor different timings. The defender must allocate a shared stock of resources across requirements that are individually manageable but jointly incompatible.

Yu-Gi-Oh! supplies a concrete example. D.D. Crow[1] can remove a card from the opponent's graveyard in response to an effect. Using it before a revival resolves can stop an immediate attack; keeping it can stop a later revival once its target is announced. Acting early consumes the resource, whereas waiting can let the original target leave the zone where Crow is effective. Section 2 gives a position in which an unknown hand card determines which timing wins.

This position leads to a question about the size of a conflict: if a family of hidden states cannot be defended with $k$ interruptions, how many states are needed to witness that failure? We address this question for a two-stage removal model. A hidden state $\theta$ specifies an urgent set $U_\theta$, a late target pool $L_\theta$, and a late-request quota $r_\theta$. The defender chooses one early removal set for all possible states and subsequently responds to observed targets. Every target costs one unit to remove.

For a finite information class $B$, write $U_B = \bigcup_{\theta \in B} U_\theta$. We show that its minimum required budget is

$$\kappa(B) = |U_B| + \max_{\theta \in B} \min\{r_\theta, |L_\theta \setminus U_B|\}, \tag{1}$$

where the maximum over the empty set is zero. The principal result is the sharp implication

$$\kappa(B) > k \quad \Longrightarrow \quad \exists C \subseteq B : \ |C| \leq k+1 \text{ and } \kappa(C) > k.$$

The proof associates urgent targets with the states that require them and, when necessary, adds one state witnessing excessive late demand. A uniform lower bound on late demand after the canonical early action sharpens the bound.

*Key words and phrases.* Hidden information, interruption timing, finite resources, Helly-type theorem, minimal obstruction, hypergraph.

[1]Japanese card name: D.D.クロウ.

The model is a two-stage robust decision problem: an initial commitment is followed by an adaptive response to uncertainty [4, 5]. Equal single-target costs yield a canonical early action and a sharp bound on conflicting states. This Helly-type bound relates global infeasibility to small infeasible subfamilies [1, 2, 3]. Minimal obstructions also describe which states can share a successful signal, as in hypergraph formulations of information compatibility [8]. The main contribution is the sharp bound on obstruction size in terms of the removal budget and the residual demand after the canonical early action. The higher-order examples further describe the structure of this feasibility system.

## 2. A graveyard-removal decision

Consider the following response window in Main Phase 1 of the attacker's turn. The defender has 4000 Life Points, an empty field, and a hand consisting of D.D. Crow and Ookazi.[2] The attacker has 800 Life Points and has activated a previously set Call of the Haunted,[3] targeting a particular copy $b_1$ of Blue-Eyes White Dragon[4] in the attacker's graveyard. That graveyard contains all three copies $b_1, b_2, b_3$ and a Blue-Eyes Ultimate Dragon[5] that was properly Fusion Summoned before being sent there. The attacker controls no monsters, and Call is the only card on either field. It is Chain Link 1, with the defender next to respond. The defender's graveyard contains no monsters. Both decks are nonempty and both Extra Decks are empty. All other cards and effects are inactive throughout the continuation.

The attacker's only hand card is either Axe of Despair[6] or Monster Reborn.[7] These two hands define the information class for the decision. The two states share a public history: exchanging the unrevealed card with its alternative in the remaining deck leaves all earlier revealed actions unchanged. The defender drew Ookazi during the current Standby Phase through a previously set Jar of Greed,[8] so it first became available during the attacker's turn. The card effects and response windows are specified in the official card database and timing guide [6, 7].

Two defensive policies illustrate the timing conflict. The *early* policy chains Crow to Call and removes $b_1$. The *reserved* policy lets Call resolve and retains Crow to remove the target of a subsequent Monster Reborn. In either policy, the defender activates Ookazi on the next turn if that turn is reached.

| Defensive policy | Hidden Axe | Hidden Reborn |
|---|---|---|
| Remove $b_1$ immediately | Defender wins | Attacker can win |
| Reserve Crow for Reborn | Attacker can win | Defender wins |

Against Axe, early removal leaves the attacker without a monster. If Call resolves instead, the revived White Dragon gains 1000 ATK and deals 4000 damage. Against Reborn, early removal permits the later revival of Ultimate Dragon, whose 4500 ATK is lethal. Reserving Crow removes Reborn's announced target before resolution and leaves at most the original 3000-ATK attacker. The defender then survives and wins with Ookazi's 800 damage. Reborn chooses its target before Crow is discarded, and the defender's graveyard initially contains no monster that could serve as an alternative target.

Other early targets do not resolve the conflict. Removing any card other than $b_1$ leaves the Axe line intact. Against Reborn, removing Ultimate Dragon still permits a second

[2]Japanese card name: 昼夜の大火事.
[3]Japanese card name: リビングデッドの呼び声.
[4]Japanese card name: 青眼の白龍.
[5]Japanese card name: 青眼の究極竜.
[6]Japanese card name: デーモンの斧.
[7]Japanese card name: 死者蘇生.
[8]Japanese card name: 強欲な瓶.

White Dragon to join $b_1$, for 6000 total attack damage; removing another card leaves Ultimate Dragon available. Consequently, with one Crow the defender has a winning policy in each fully specified state, but no single policy guarantees victory in both. Two Crows suffice: remove $b_1$ first and reserve the second for a later revival from the attacker's graveyard. Reborn can alternatively summon the defender's discarded Crow, whose 100 ATK is nonlethal.

The position illustrates competition between urgent and deferred uses of the same resource. In the model below, each unanswered target causes failure and each removal costs one unit. This formulation isolates the resource interaction from the attack-damage calculation in the card position.

## 3. The removal model and its exact threshold

Let $X$ be a set of targets and $\Theta$ a set of hidden states. For each $\theta \in \Theta$, let $U_\theta, L_\theta \subseteq X$ be finite sets and let $r_\theta \in \mathbb{N}$, where $0 \in \mathbb{N}$. The defender knows a finite information class $B \subseteq \Theta$ and has $k \in \mathbb{N}$ identical removal units. For nonempty $B$, a state $\theta \in B$ is fixed throughout play and hidden from the defender. Every target in the specified sets is available for early removal; the uncertainty concerns which requirements will apply.

Before an urgent deadline, the defender chooses a finite set $E \subseteq X$ and spends $|E|$ units. The same $E$ is used in every state in $B$. In the actual state $\theta$, every urgent target must belong to $E$. After the deadline, the adversary may request up to $r_\theta$ distinct targets from $L_\theta \setminus E$, one at a time. The defender observes each requested target and spends one unit to remove it. An unanswered urgent or late target causes failure. Removed targets remain removed, and spent units are not replenished. Figure 1 shows the timing.

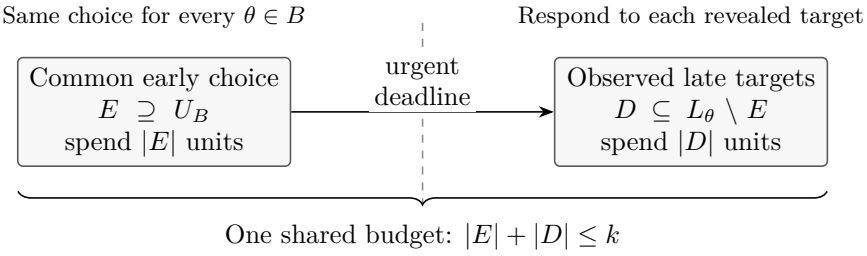



Figure 1. The early choice is common to the information class. Later responses use the identities of the requested targets. Both stages draw on the same budget.

**Definition 3.1.** An early set $E$ is *safe for $B$ at budget $k$* if $|E| \le k$ and, for every $\theta \in B$,

$$U_\theta \subseteq E, \qquad |E| + |D| \le k \quad \text{for every } D \subseteq L_\theta \setminus E \text{ with } |D| \le r_\theta. \tag{2}$$

The family $B$ is *feasible at budget $k$* if it has a safe early set. Its minimum feasible budget is denoted by $\kappa(B)$.

A finite feasible budget always exists: remove $U_B$ early and retain enough units for the largest remaining late pool. In particular, $\kappa(\varnothing) = 0$.

The safety definition characterizes successful sequential defense. If it holds, spending one unit on each observed request succeeds against every sequence, including adaptively chosen sequences. Conversely, early expenditure exceeding $k$ is inadmissible, and a missed urgent target causes failure at the deadline. In the remaining case, failure of safety provides a state and a legal set $D$ with $|E| + |D| > k$. Requesting its elements in any order forces an unanswered request. Observing the requested targets therefore suffices for a successful response policy.

**Lemma 3.2.** *For fixed $\theta$ and $E$, the early expenditure plus the largest admissible late demand is*

$$c_\theta(E) = |E| + \min\{r_\theta, |L_\theta \setminus E|\} = \min\{|E| + r_\theta, |E \cup L_\theta|\}. \tag{3}$$

*If $E \subseteq F$, then $c_\theta(E) \le c_\theta(F)$.*

*Proof.* The adversary can request exactly $\min\{r_\theta, |L_\theta \setminus E|\}$ remaining targets and cannot request more. The second equality follows from $|E \cup L_\theta| = |E| + |L_\theta \setminus E|$. Both arguments of the final minimum are nondecreasing as $E$ grows, which proves monotonicity. □

**Theorem 3.3** (Exact threshold)**.** *For every finite $B \subseteq \Theta$, formula* (1) *holds. In particular, $B$ is feasible at budget $k$ if and only if $\kappa(B) \le k$. Whenever $B$ is feasible at budget $k$, the early set $E = U_B$ is safe at that budget.*

*Proof.* Every safe early set contains $U_B$. By Lemma 3.2, replacing such a set by $U_B$ weakly decreases the total cost in every state and also weakly decreases early expenditure. Thus $U_B$ is an optimal early set. Taking the maximum late demand over $B$ gives the formula. For the empty family, both terms are zero. □

**Corollary 3.4.** *If $C \subseteq B$, then $\kappa(C) \le \kappa(B)$.*

*Proof.* Every early set safe for $B$ is safe for $C$ at the same budget. □

The canonical action has a simple interpretation: an extra early removal can save at most the one unit it costs later. Removing every possible urgent target and deferring all other removals therefore attains the minimum worst-case budget.

## 4. Sharp bounds on infeasible subfamilies

**Lemma 4.1** (Selecting witnesses)**.** *If $S \subseteq U_B$, then there is a subfamily $C \subseteq B$ such that $S \subseteq U_C$ and $|C| \le |S|$.*

*Proof.* For each $x \in S$, choose a state $\theta_x \in B$ with $x \in U_{\theta_x}$, and take $C = \{\theta_x : x \in S\}$. □

**Theorem 4.2** (Sharp obstruction bound)**.** *For every finite $B \subseteq \Theta$ and every $k \in \mathbb{N}$,*

$$\kappa(B) > k \iff \exists C \subseteq B : \ |C| \le k+1 \text{ and } \kappa(C) > k. \tag{4}$$

*For every $k$, the bound $k+1$ is attained by an inclusion-minimal infeasible family.*

*Proof.* The reverse implication follows from Corollary 3.4. For the forward implication, first suppose $|U_B| > k$. Choose $S \subseteq U_B$ of size $k+1$ and apply Lemma 4.1. The resulting $C$ has at most $k+1$ states and requires at least $k+1$ early removals, so $\kappa(C) > k$.

Now suppose $|U_B| \le k$. By Theorem 3.3, some $\theta_* \in B$ satisfies $c_{\theta_*}(U_B) > k$. Apply Lemma 4.1 to $S = U_B$, obtaining a family $C_0$ of at most $|U_B|$ states whose urgent union is $U_B$. Set $C = C_0 \cup \{\theta_*\}$. Then $U_C = U_B$ and $\theta_* \in C$, so the same excessive late cost persists in $C$. Moreover, $|C| \le |U_B| + 1 \le k+1$.

For sharpness, take $k+1$ states indexed by $i$, with $U_i = \{i\}$, $L_i = \varnothing$, and $r_i = 0$. Every subfamily $C$ has $\kappa(C) = |C|$. Thus the full family is infeasible and every proper subfamily is feasible at budget $k$. □

Equivalently, feasibility of all subfamilies of size at most $k+1$ implies feasibility of the whole family. The bound is independent of the number of available states and target labels.

**Definition 4.3.** A number $d \in \mathbb{N}$ is a *uniform late reserve for $B$* if

$$d \le \min\{r_\theta, |L_\theta \setminus U_B|\} \qquad \text{for every } \theta \in B. \tag{5}$$

The reserve is measured at the canonical early set $E = U_B$. Removing additional late targets early can reduce later demand, but Lemma 3.2 shows that it cannot reduce the combined expenditure. If $C \subseteq B$, then $U_C \subseteq U_B$, so (5) also holds for each $\theta \in C$ with $U_C$ in place of $U_B$.

**Theorem 4.4** (Reserve refinement)**.** *Suppose $d$ is a uniform late reserve for $B$. If $\kappa(B) > k$, then there is a subfamily $C \subseteq B$ such that*

$$\kappa(C) > k, \qquad |C| \leq 1 + \max\{k - d, 0\}.$$

*The bound is sharp for every pair $k, d \in \mathbb{N}$.*

*Proof.* An infeasible family is nonempty. If $d > k$, any singleton from $B$ already has late demand exceeding $k$ and supplies the required witness. Suppose henceforth that $d \leq k$.

If $|U_B| + d > k$, choose $k - d + 1$ urgent targets and select their witness states by Lemma 4.1. The resulting nonempty family $C$ satisfies $|C| \leq k-d+1$ and $|U_C| \geq k-d+1$. The inherited reserve gives $\kappa(C) \geq |U_C| + d \geq k + 1$.

If $|U_B| + d \leq k$, use the second construction in the proof of Theorem 4.2. It selects all urgent witnesses together with one state of excessive late cost and has size at most $|U_B| + 1 \leq k - d + 1$.

For sharpness when $d \leq k$, let $n = k - d + 1$. Take $n$ distinct urgent singletons and a common late pool $P$ of size $d$ disjoint from all urgent targets, with quota $d$ in every state. Each nonempty subfamily $C$ has $\kappa(C) = |C| + d$. The full family costs $k + 1$ and every proper subfamily costs at most $k$. When $d > k$, take one state with empty urgent set, a late pool of size $d$, and quota $d$. Its threshold is $d$, and its only proper subfamily is empty and has threshold zero. □

For $d \leq k$, the canonical policy must retain capacity for at least $d$ late requests in every state, leaving at most $k - d$ units for its urgent union. One additional state can witness the failure of a common defense.

**Proposition 4.5** (Private urgent targets)**.** *Let $C$ be an inclusion-minimal family infeasible at budget $k$, with $|C| \geq 2$. All but at most one state $\theta \in C$ have a target in $U_\theta \setminus U_{C\setminus\{\theta\}}$.*

*Proof.* If $|U_C| > k$, deleting a state with no such target leaves the urgent union unchanged and hence leaves an infeasible proper subfamily. Thus every state has a private urgent target in this case. Otherwise, fix $\theta_* \in C$ with $c_{\theta_*}(U_C) > k$. Deleting a state other than $\theta_*$ that has no private urgent target preserves both $U_C$ and this late-cost witness, again contradicting minimality. □

This proposition gives a second counting proof of both bounds. Delete the possible exceptional state, or any state if there is no exception. The remaining family is nonempty and feasible, and contains at least $|C| - 1$ distinct urgent targets. Under a uniform reserve $d$ it therefore satisfies $|C| - 1 + d \leq k$. Singletons satisfy the stated bounds directly.

## 5. Higher-order conflicts and the role of single-target actions

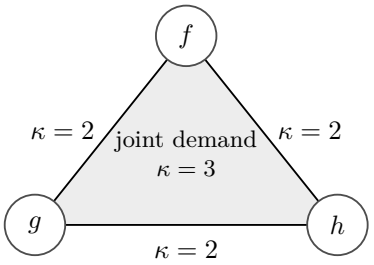
5.1. **Pairwise feasibility and increasing marginal cost.** Take three states with the following data:

| State | Urgent set | Late pool | Quota |
|---|---|---|---|
| $f$ | $\{a\}$ | $\{b\}$ | 1 |
| $g$ | $\{b\}$ | $\varnothing$ | 0 |
| $h$ | $\varnothing$ | $\{c\}$ | 1 |

The exact threshold gives

$$\kappa(\{f\}) = \kappa(\{f,g\}) = \kappa(\{f,h\}) = \kappa(\{g,h\}) = 2, \quad \kappa(\{f,g,h\}) = 3. \tag{6}$$

Every pair can be defended with two units; the triple cannot (Figure 2). For $\{f,h\}$, removing $a$ early leaves one unit for whichever late target is requested. Adding $g$ makes $b$ an urgent target. If the actual state is $h$, a further unit is still needed for $c$.




$f$ : $U_f = \{a\},\ L_f = \{b\}$
$g$ : $U_g = \{b\},\ L_g = \varnothing$
$h$ : $U_h = \varnothing,\ L_h = \{c\}$

FIGURE 2. A conflict involving three states at budget two. Each edge denotes a feasible pair. The shaded triple is a minimal infeasible family. The late quotas of $f$ and $h$ are one; the quota of $g$ is zero.

**Proposition 5.1.** *The threshold function need not be submodular and need not satisfy LP-type locality.*

*Proof.* Let $A = \{f\}$ and $B = \{f,g\}$ in (6). The marginal cost of adding $h$ is zero on $A$ and one on $B$, violating the diminishing returns inequality for submodularity.

LP-type locality requires that, whenever $A \subseteq B$ and $\kappa(A) = \kappa(B)$, a state that strictly raises the value of $B$ also strictly raises the value of $A$ [1, 2]. Here $\kappa(A) = \kappa(B) = 2$, whereas adding $h$ raises only the value of $B$. □

Short infeasibility certificates and locality are distinct properties in the general theory [2]. The example identifies their separation within the present resource objective: a continuation can become costly through its interaction with another possible continuation.

5.2. **Resource budget and constraint-set cardinality.** For comparison, a family of sets, each of cardinality at most $k$, has Helly number at most $k+1$ [3, Theorem 7]. Our parameter counts removal units. A state's feasible early plans can form a much larger set: at budget two, a state with urgent set $\{a\}$ and no late requests permits every plan $\{a,x\}$.

The distinction persists under alternative exact intersection representations. Take four states with distinct urgent singletons, no late requests, and budget two. Every pair is feasible and every triple is infeasible. Suppose sets $S_1,\dots,S_4$ represent this feasibility pattern by nonempty intersections. Choose $x_j \in S_1 \cap S_j$ for $j = 2,3,4$. These three points are distinct, since an equality would give a point in three sets. Hence $|S_1| \geq 3$. Thus this budget-two example requires constraint sets larger than two in any exact intersection representation.

5.3. **Actions that remove several targets.**

**Proposition 5.2.** *When one unit may remove several targets, minimal infeasible families can have arbitrarily large cardinality at budget one.*

*Proof.* Fix $n \geq 2$ and take $n$ states, with urgent set $\{i\}$ in state $i$ and no late requests. For each $j$, provide an action $a_j$ that costs one unit and removes all targets except $j$. Every proper subfamily omits some state $j$ and is defended by $a_j$. No available action defends the full family. Its cardinality is $n$, as required. □

The unit cost of an individual target is therefore the mechanism behind Theorem 4.2. It makes every additional indispensable urgent requirement consume a separate part of the budget.

## 6. Signals and minimal-obstruction hypergraphs

Fix a finite state space $\Omega \subseteq \Theta$ and budget $k$. Define a hypergraph $\mathcal{H}_k$ on $\Omega$ whose edges are the inclusion-minimal families $C$ with $\kappa(C) > k$. By Theorem 4.2, every edge has size at most $k+1$. Under a uniform late reserve $d$ for $\Omega$, every edge has size at most $1 + \max\{k-d, 0\}$.

Suppose an informed sender supplies a deterministic signal $s : \Omega \to M$ with finite message alphabet $M$ before the early choice. We count the messages in $s(\Omega)$; the empty state space uses zero messages and has weak chromatic number zero. The defender then selects an early set using the received message. The message class $s^{-1}(m)$ is exactly the set of states that this choice must jointly defend.

**Proposition 6.1.** *A signal permits guaranteed defense at budget $k$ if and only if no edge of $\mathcal{H}_k$ is monochromatic. If every singleton is feasible, the minimum number of messages equals the weak chromatic number of $\mathcal{H}_k$.*

*Proof.* A message class is feasible precisely when it contains no minimal infeasible subfamily. Indeed, an infeasible finite class contains such a subfamily, and any class containing one is infeasible by monotonicity. Thus every message class is feasible exactly when no hyperedge lies inside one class. This is the definition of a weak proper coloring. If every singleton is feasible, assigning a separate message to each state gives a proper coloring, so both minima exist and coincide. □

The use of common successful decisions to characterize information classes is shared with hypergraph approaches to coding for function computation [8]. Here the resource bounds control the sizes of the minimal forbidden classes. In the example of (6), the sole minimal obstruction is the three-state set: any two states may share a message, but all three may not. Pairwise conflict graphs therefore omit an interaction that is decisive for successful signaling.

## 7. Formalization and further questions

The accompanying Lean sources formalize the exact threshold and its minimum-budget interpretation (Theorem 3.3), monotonicity, the witness-selection lemma, and both sharp obstruction bounds (Theorems 4.2 and 4.4), including the singleton case $d > k$. They also formalize the three-state values and the counterexamples in Propositions 5.1 and 5.2. Safety universally quantifies over admissible late target sets. The proofs derive maximum late demand, establish the optimality of the urgent union, and select target witnesses. The sequential interpretation, private-target argument, intersection representation, and signal-coloring correspondence have the direct proofs given above.

One direction is to characterize the minimal-obstruction hypergraphs of removal models. Such a characterization would describe the information partitions compatible with a given resource budget. A second direction allows heterogeneous costs or multiple urgent deadlines; the monotonicity identity (3) identifies an exchange property to examine in those settings. Card effects that alter target accessibility or remove several targets also provide concrete settings for studying how interruption granularity changes the order of hidden-state conflicts.

**Declaration of AI assistance.** GPT-6 Astra assisted with the mathematical derivations in this paper.

## References


[1] Nina Amenta, *Helly-type theorems and generalized linear programming*, Discrete & Computational Geometry **12** (1994), 241–261. https://www.cs.ucdavis.edu/~amenta/pubs/helly.pdf.

[2] Annamaria Beatrice Amenta, *Helly Theorems and Generalized Linear Programming*, Ph.D. thesis, University of California, Berkeley, 1993. https://www.cs.ucdavis.edu/~amenta/pubs/ninathesis.pdf.

[3] Jean Cardinal, Hiro Ito, Matias Korman, and Stefan Langerman, *Helly numbers of polyominoes*, Proceedings of the 23rd Canadian Conference on Computational Geometry, 2011. https://cccg.ca/proceedings/2011/papers/paper104.pdf.

[4] Marc Goerigk, Stefan Lendl, and Lasse Wulf, *Two-stage robust optimization problems with two-stage uncertainty*, arXiv:2104.03043, 2021. https://arxiv.org/abs/2104.03043.

[5] Adam Kasperski and Paweł Zieliński, *Robust recoverable and two-stage selection problems*, arXiv:1505.06893, 2015. https://arxiv.org/abs/1505.06893.

[6] Konami Digital Entertainment, *Yu-Gi-Oh! card database*, English and Japanese card entries. https://www.db.yugioh-card.com/yugiohdb/.

[7] Konami Digital Entertainment, *Fast effect timing*. https://www.yugioh-card.com/en/play/fast-effect-timing/.

[8] Deheng Yuan, Tao Guo, Bo Bai, and Wei Han, *Lossy computing with side information via multi-hypergraphs*, arXiv:2205.08262, 2022. https://arxiv.org/abs/2205.08262.